

Planetary scientist and space systems engineer with experience at ESA and NASA JPL supporting planetary mission science, observation planning, scientific data analysis and research software development. Background in lunar and planetary geophysics, scientific instrumentation concepts, mission science and multidisciplinary collaboration across science and engineering teams. Current research investigates lunar interior structure and composition to support the interpretation of future geophysical observations and planetary mission data.

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EXPERIENCE

NASA Jet Propulsion Laboratory (NASA JPL)

- **2026 | Visiting Student Researcher | United States (6 months)**

- Develop thermodynamic and geophysical models of the Moon and icy moons to test interior-structure and composition hypotheses against mission-relevant observables.
- Define and compare bulk-silicate Moon composition scenarios and assess their implications for lunar mantle structure and future geophysical measurements.
- Develop and validate reproducible scientific software workflows, documenting model assumptions, limitations and scientific results.

European Space Agency (ESA)

- **2023-2025 | Young Graduate Trainee: BepiColombo & JUICE missions | Netherlands (2 years)**

- Developed SolarConflux, an open-source Python tool supporting coordinated observation planning across more than 20 space missions.
- Supported observation-planning studies and multi-mission scientific analysis for BepiColombo and other solar-system missions.
- Developed a centralised scientific data and knowledge platform supporting JUICE mission preparation and access to mission information across science and operations teams.
- Coordinated scientific and technical content with multidisciplinary stakeholders, improving information traceability and knowledge sharing.

- **2023 - 2024 | Embedded Software Engineer: YPSat ESA mission | Netherlands (1 year, part-time)**

- Contributed to embedded software for the spacecraft Wake-Up Subsystem within mission-level interface and launch-sequence constraints.
- Supported software integration, technical coordination and documentation for an operational payload flown on Ariane 6.

- **2022 | Intern: BepiColombo ESA mission | Spain (5 months)**

- Supported science planning and observation preparation for the BepiColombo mission.
- Processed and analysed MESSENGER datasets to support scientific operations and mission activities.
- Produced technical documentation and scientific analyses for mission planning and coordination.

Tohoku University

- **2023 | Planetary Science Research Intern: MMX mission context | Japan (4 months)**

- Investigated origin scenarios for Phobos and the Martian moons through spectral data analysis and analogue rock-sample studies.
- Conducted laboratory experiments using FTIR spectroscopy to characterise Phobos analogue samples and analyse mineralogical and compositional signatures.
- Interpreted experimental and spectral results in the context of the MMX mission's scientific objectives.

Atmospheres and Space Observations Laboratory (LATMOS)

- **2022 | Research Student: Ceres mission concept | France (1 year)**

- Performed preliminary payload sizing and instrument trade-off studies for a 3U CubeSat radar sounder, translating subsurface-science requirements into payload and measurement constraints.
- Defined scientific objectives and a preliminary measurement concept for investigating Ceres' subsurface structure and potential cryovolcanic features.
- Assessed expected scientific applications and instrument performance requirements related to geomorphology, subsurface layering and cryovolcanism.

PROJECTS & OPEN-SOURCE SCIENTIFIC SOFTWARE

Sustainable Lunar Exploration Architecture Team study of ISRU, logistics and system architectures enabling sustained human and robotic operations on the Moon.

PlanetProfile Applied and extended the open-source PlanetProfile framework (JPL) to model thermodynamic and geophysical properties of icy worlds and the lunar interior.

Planetary EOS Lab Python framework streamlining Perple_X workflows for planetary interior modelling and integration with PlanetProfile.

SolarConflux Python tool identifying coordinated observation opportunities across heliophysics missions for ESA mission planning.

SKILLS

Scientific Instrumentation and Mission Science

Measurement concepts, preliminary payload studies, observation planning, science requirements, payload data analysis, laboratory spectroscopy

Lunar and Planetary Science

Lunar geophysics, planetary interiors, geochemistry, spectroscopy, surface and subsurface processes, mission science

Scientific Computing and Data Analysis

Python, numerical modelling, scientific data processing, model validation, Git/GitHub, reproducible research workflows

Space Systems and Coordination

Mission design, embedded software, science operations support, technical documentation, knowledge management, multidisciplinary coordination

CONFERENCES & SCIENTIFIC MEETINGS

2026

AAS Division for Planetary Sciences (DPS) Meeting | [Submitted](#) | Lunar interior modelling and bulk-silicate mantle compositions.

2025

BepiColombo Science Working Team | [Talk](#) | SolarConflux: Coordinated heliophysics observation opportunities across multiple spacecraft using mission geometry analysis.

2024

Euromet Science Congress (EPSC) | [Poster](#) | JUICE Mission Preparation: Science operations, observation planning and mission coordination for the Jupiter Icy Moons Explorer.

Association of Space Explorers (ASE) Planetary Congress | [Talk](#) | The Future of Space Exploration: Outreach presentation on future space missions, lunar exploration and the role of the next generation of space professionals.

JUICE Science Working Team | [Poster](#) | JUICE Mission Preparation: Science operations, observation planning and mission coordination for the Jupiter Icy Moons Explorer.

EDUCATION

TAS Astro Advanced Master

ISAE-SUPAERO | 2025 - 2026

Space Studies Program (SSP)

International Space University | 2025

MSc Planetary Sciences and Space Exploration

Université Paris Saclay | 2022 - 2023

MSc Space Science and Technology

Observatoire de Paris | 2021 - 2022

Bachelor's degree in Fundamental Physics

Clermont Auvergne University | 2019 - 2021

CERTIFICATES

Machine Learning Specialization

Stanford University | 2024

Mathematics for Machine Learning

Imperial College London | 2024

LANGUAGES

French (Native)

English (Fluent - 98 TOEFL iBT)

Japanese (Beginner)

Spanish (Intermediate)